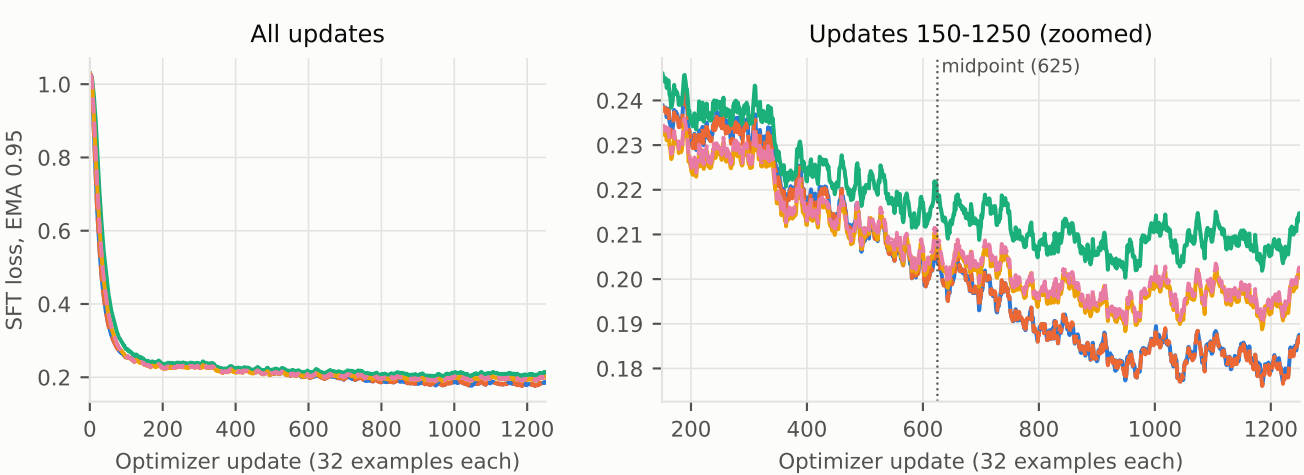

Full FT
L2-SP
LoRA r=4
LoRA r=16
OPLoRA r=16
All updates
SFT loss, EMA 0.95
1.0
0.8
0.6
0.4
0.2
0
200
400
600
800
1000
1200
Optimizer update (32 examples each)
Updates 150-1250 (zoomed)
midpoint (625)
0.24
0.23
0.22
0.21
0.20
0.19
0.18
200
400
600
800
1000
1200
Optimizer update (32 examples each)